Bantu African dataset maps

2025-11-04

```
res_dir <- "../results/s10_data_source_map/"
```

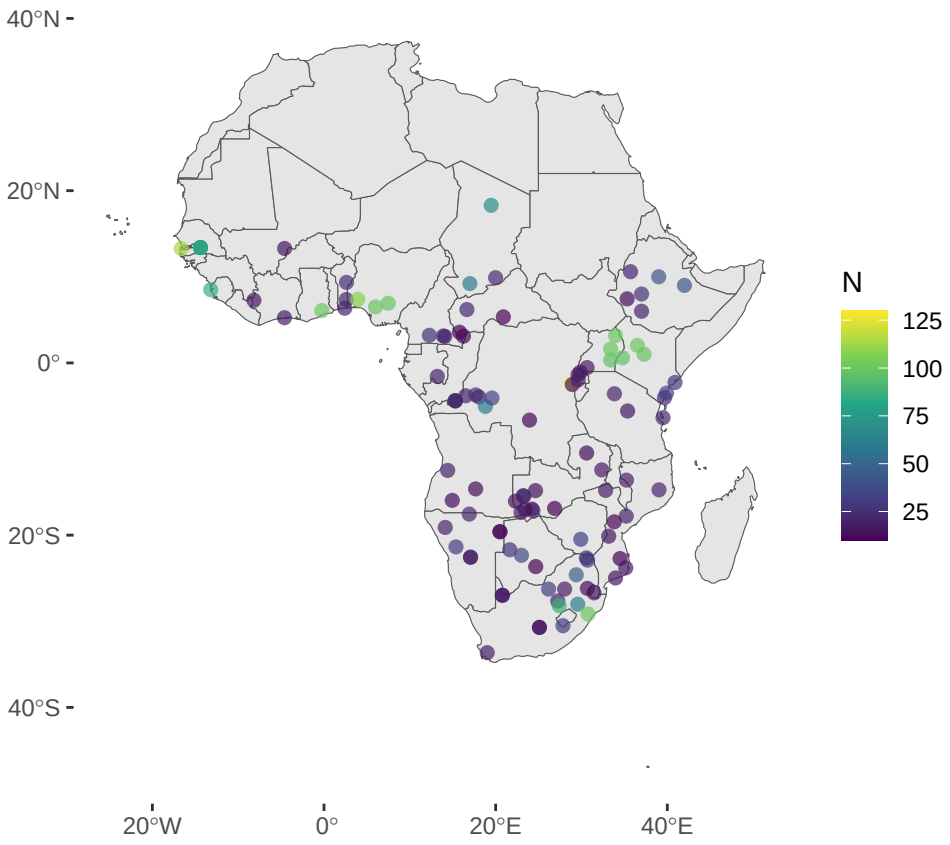
Plot metadata

```
dat2 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 2))
```

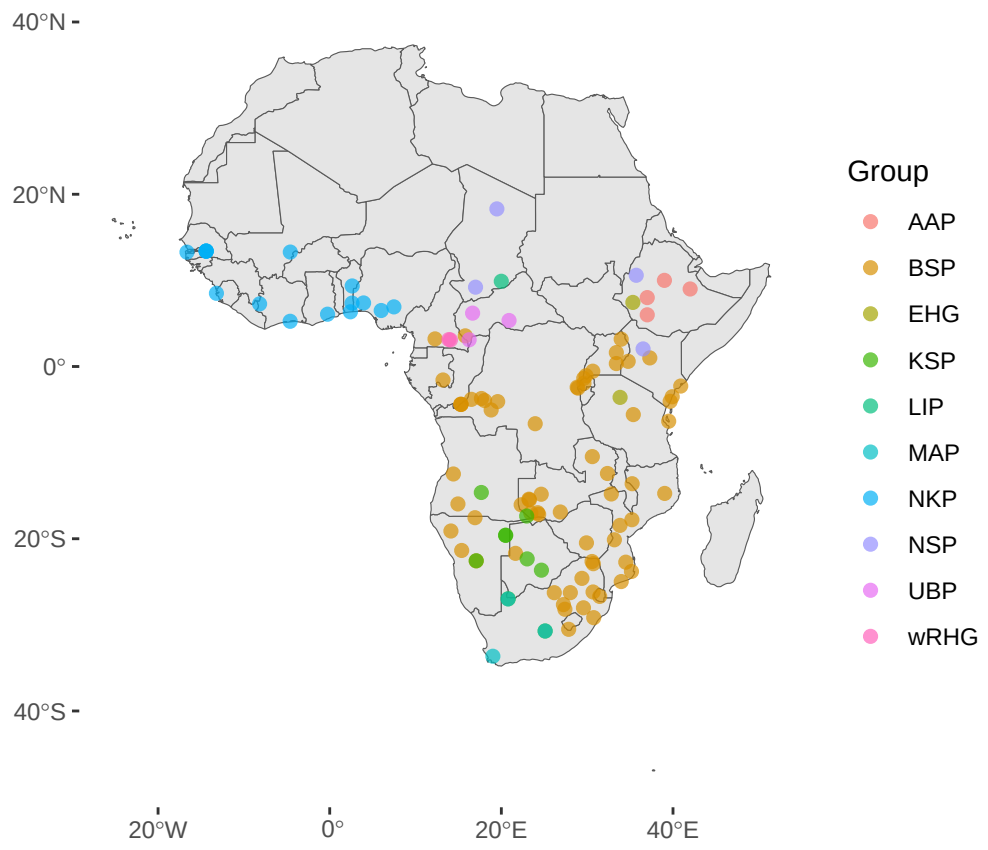
```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
```

```
colnames(dat2) <- dat2[1, ]
dat2 <- dat2[-1, ]
# rename the 'Only-Africa dataset' column
colnames(dat2)[which(colnames(dat2) == "Only-Africa dataset")] <- "Africa"
# include only entries in Africa
dat2 <- dat2 %>%
  filter(Africa == "Yes")
# remove rows with missing lat/lon
dat2 <- dat2 %>%
  filter(!is.na(Latitude) & !is.na(Longitude))
# transform to sf object
dat2 <- st_as_sf(dat2, coords = c("Longitude", "Latitude"), crs = 4326)
# make sample size numeric
dat2$N <- as.numeric(dat2$N)
# create plot of Africa
africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")
# plot by sample size
p2 <- ggplot() +
  geom_sf(data = africa) +
```

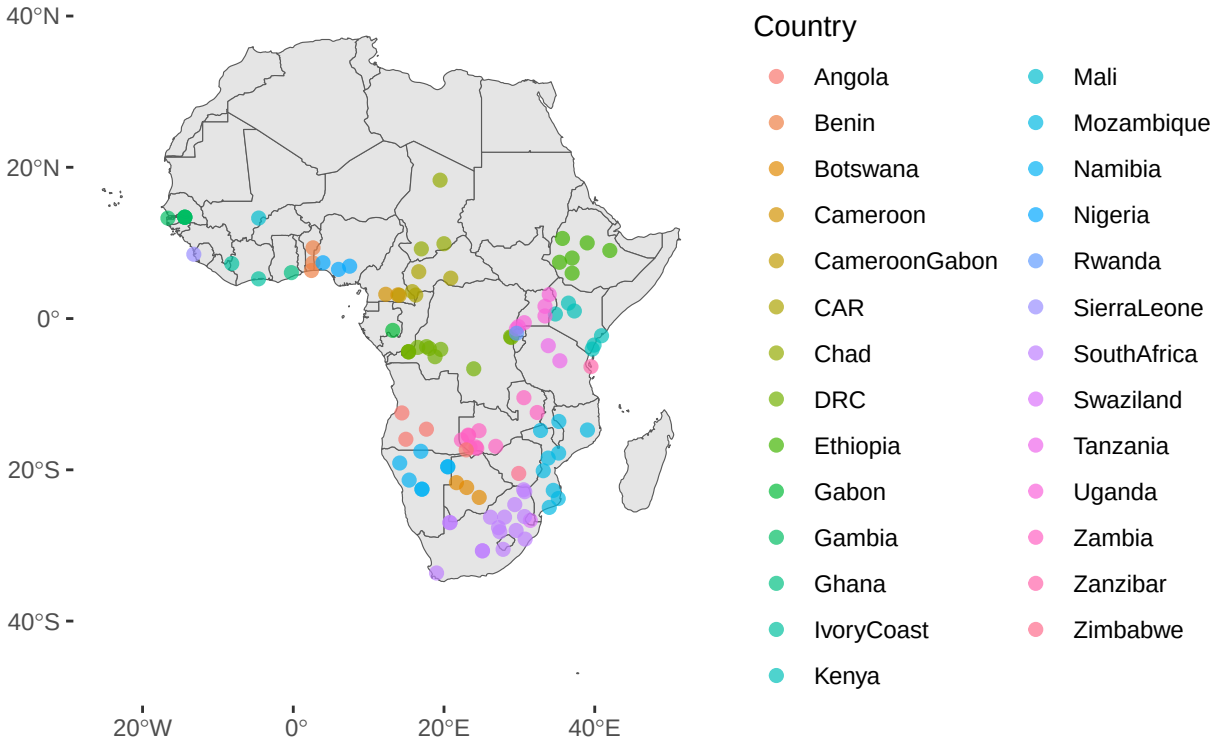
```
  geom_sf(data = dat2, aes(color = N), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA)) +
  scale_color_viridis_c()
p2
```



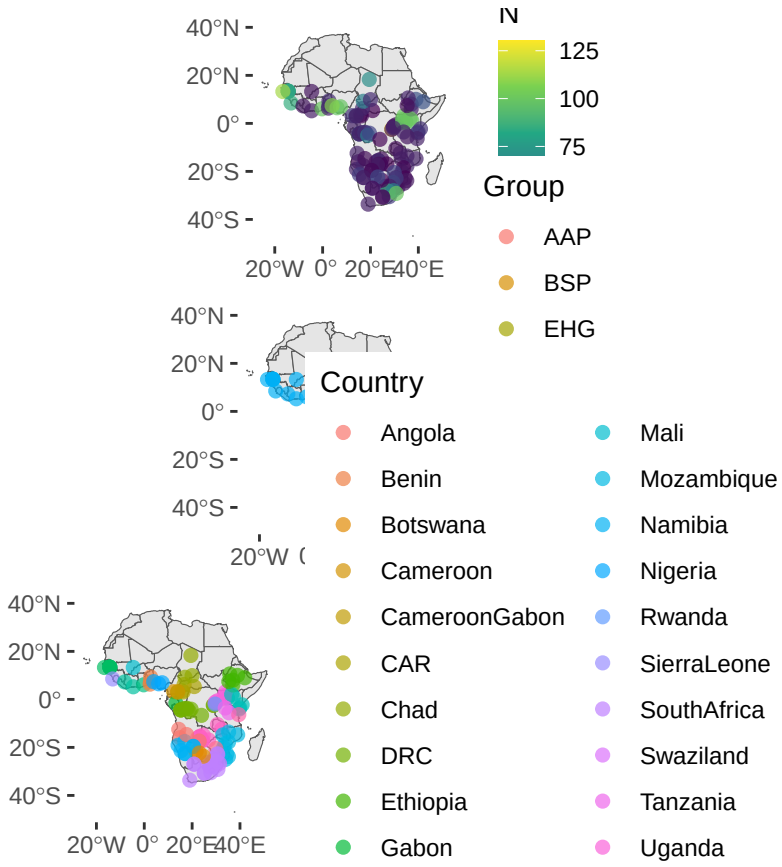

```
# plot by group
p3 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat2, aes(color = Group), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA))
p3
```




```
# plot by country
p4 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat2, aes(color = Country), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA))
p4
```




```
grid.arrange(p2, p3, p4, ncol = 1)
```




```
#
```

```
NEO_full <- gt_load("../data/intermediate/rbind_data/NEO_full.gt")
```

```
## your gen_tibble was in an old format, and it has now been updated
## note that the old RDS was overwritten
```

```
length(unique(NEO_full$population))
```

```
## [1] 164
```

```
length(unique(dat2$`Population label`))
```

```
## [1] 111
```

```
dat1 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 1))
```

```
## New names:
## * `` -> `...1`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
```

```
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
```

```
colnames(dat1) <- dat1[1, ]
dat1 <- dat1[-1, ]

NEO_full$dataset <- dat1[match(NEO_full$id, dat1$`ID Lab`),"EGA Dataset"]
```

```
dat2 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 2))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
```

```
colnames(dat2) <- dat2[1, ]
dat2 <- dat2[-1, ]
# rename the 'Only-Africa dataset' column
colnames(dat2)[which(colnames(dat2) == "Only-Africa dataset")] <- "Africa"
# include only entries in Africa
dat2 <- dat2 %>%
  filter(Africa == "Yes")
# remove rows with missing lat/lon
dat2 <- dat2 %>%
  filter(!is.na(Latitude) & !is.na(Longitude))
# subset to just `Population label` and Latitude and Longitude
dat2 <- dat2[, c("Population label", "Latitude", "Longitude")]
# rename `Population label` to Population
colnames(dat2)[which(colnames(dat2) == "Population label")] <- "Population"

dat3 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 3))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
```

```r
colnames(dat3) <- dat3[1, ]
dat3 <- dat3[-1, ]
dat3 <- dat3[, c("Population", "Latitude", "Longitude")]

dat4 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 4))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...16`
## * `` -> `...17`
## * `` -> `...18`
## * `` -> `...19`
## * `` -> `...20`
## * `` -> `...21`
```

```r
colnames(dat4) <- dat4[1, ]
dat4 <- dat4[-1, ]
dat4 <- dat4[, c("Population", "Latitude", "Longitude")]

dat12 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 12))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
```

```r
colnames(dat12) <- dat12[1, ]
dat12 <- dat12[-1, ]

dat10 <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 10))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
```

```
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
```

```
colnames(dat10) <- dat10[1, ]
dat10 <- dat10[-1, ]
dat10 <- dat10[, c("Populations", "Latitude", "Longitude")]
colnames(dat10)[which(colnames(dat10) == "Populations")] <- "Population"
```

```
all_pops <- rbind(dat2, dat3, dat4, dat10)
dups <- duplicated(all_pops)
# remove duplicates
all_pops <- all_pops[!dups, ]
```

Unfortunately, coordinates are missing for a lot of populations in our full dataset. In some cases the mismatch is a naming issue

eg

Namibia_DamaraKSP in dat$`Population label` is probably "Namibia_Damara" in NEO_full

There are coords for Sotho, but in NEO_full there are southern and northern Sotho

SouthAfrica_Coloured in NEO_full is probably SouthAfrica_ColouredAskham in dat$`Population label` - both have 12 individuals

```
all_pops$Population <- ifelse(all_pops$Population == "Tanzania_TanzaniaMixed",
  "Tanzania_Mixed",
  all_pops$Population
)

# rename DRC_Bangubangu to DRC_BanguBangu in NEO_full$population
NEO_full$population <- ifelse(NEO_full$population == "DRC_Bangubangu",
  "DRC_BanguBangu",
  NEO_full$population
)


setdiff(NEO_full$population, all_pops$Population)
```

```
##  [1] "Botswana_Kalanga"              "CAR_Nzakara"
##  [3] "CAR_Zande"                     "SouthAfrica_ColouredWellington"
##  [5] "SouthAfrica_Swazi"             "SouthAfrica_ColouredColesberg"
##  [7] "Namibia_Damara"                "Botswana_Kgalagadi"
##  [9] "SouthAfrica_SothoSouthern"     "SouthAfrica_Hlubi"
## [11] "SouthAfrica_Mputhi"            "Mozambique_Nhungwe"
## [13] "Mozambique_Tsonga"             "Mozambique_Ronga"
## [15] "Mozambique_Chuwabu"            "Mozambique_TongaMoz"
## [17] "SouthAfrica_Ndebele"           "SouthAfrica_SothoNorthern"
```

```
##  [19] "Botswana_TswanaBot"            "Lesotho_SothoSouthern"
##  [21] "SouthAfrica_Soto"              "DRC_Yombe"
##  [23] "DRC_Ngbandi"                   "DRC_Boo"
##  [25] "DRC_Mboma"                     "DRC_Ndibu"
##  [27] "DRC_MongoNkundo"               "DRC_Ntandu"
##  [29] "DRC_Mbunda"                    "DRC_Ngombe"
##  [31] "DRC_Tetela"                    "DRC_Sakata"
##  [33] "DRC_Nandi"                     "DRC_Ruanda"
##  [35] "DRC_Yaka"                      "DRC_Hendo"
##  [37] "DRC_Haavu"                     "DRC_BanguBangu"
##  [39] "DRC_Kanyoka"                   "DRC_Ciokwe"
##  [41] "DRC_Suku"                      "DRC_Kusu"
##  [43] "DRC_Songe"                     "DRC_Sengele"
##  [45] "DRC_Kango"                     "DRC_Kuba"
##  [47] "DRC_Zimba"                     "DRC_Lunda"
##  [49] "DRC_Binza"                     "Uganda_Konzo"
##  [51] "Zambia_Kaonde"                 "Zambia_ShonaZam"
##  [53] "Zambia_Tumbuka"                "Tanzania_TanzaniaMixed"
##  [55] "Zambia_Ngoni"                  "Zambia_Nsenga"
##  [57] "SouthAfrica_Coloured"          "Angola_Vasekele"
##  [59] "Angola_Sekele"                 "Namibia_Kgalagadi"
##  [61] "DRC_Luozi"                     "DRC_Nienge"
##  [63] "DRC_Bandundu"                  "Zambia_Tswana"
##  [65] "Zambia_Kalanga"                "Zambia_Kgalagadi"
##  [67] "Zambia_Mbukushu"               "Zambia_Kavango"
##  [69] "Zambia_Kwangari"               "Zambia_Luyana"
##  [71] "Zambia_Shona"                  "Zambia_Shanjo"
##  [73] "DRC_Wonk"                      "DRC_Nsong"
##  [75] "DRC_Nzadi"                     "DRC_Hungan"
##  [77] "DRC_Nsambaan"                  "DRC_Lele"
##  [79] "DRC_MbweneNgong"               "DRC_Mpiin"
##  [81] "DRC_LubaKasai"                 "DRC_Ngong"
##  [83] "DRC_Mpur"                      "DRC_Cokwe"
##  [85] "DRC_Mpe"                       "DRC_NunuBobangi"
##  [87] "DRC_NunuMushie"                "DRC_Ngbaka"
##  [89] "DRC_Balobo"                    "DRC_Mbudza"
##  [91] "DRC_Mongo"                     "DRC_Nande"
##  [93] "DRC_Lulua"                     "DRC_Binji"
##  [95] "DRC_Ruund"                     "DRC_LubaKatanga"
##  [97] "DRC_Mangbetu"                  "DRC_Mbaama"
##  [99] "DRC_Myene"                     "DRC_Nzebi"
## [101] "DRC_Teke"                      "DRC_Fang"
## [103] "DRC_RundiHutu"                 "DRC_Lika"
## [105] "DRC_Luba"                      "SouthAfrica_Shona"
```

```r
# add coordinates to NEO_full based on matching population labels
NEO_full_na_omit <- NEO_full %>%
  filter(population %in% all_pops$Population)

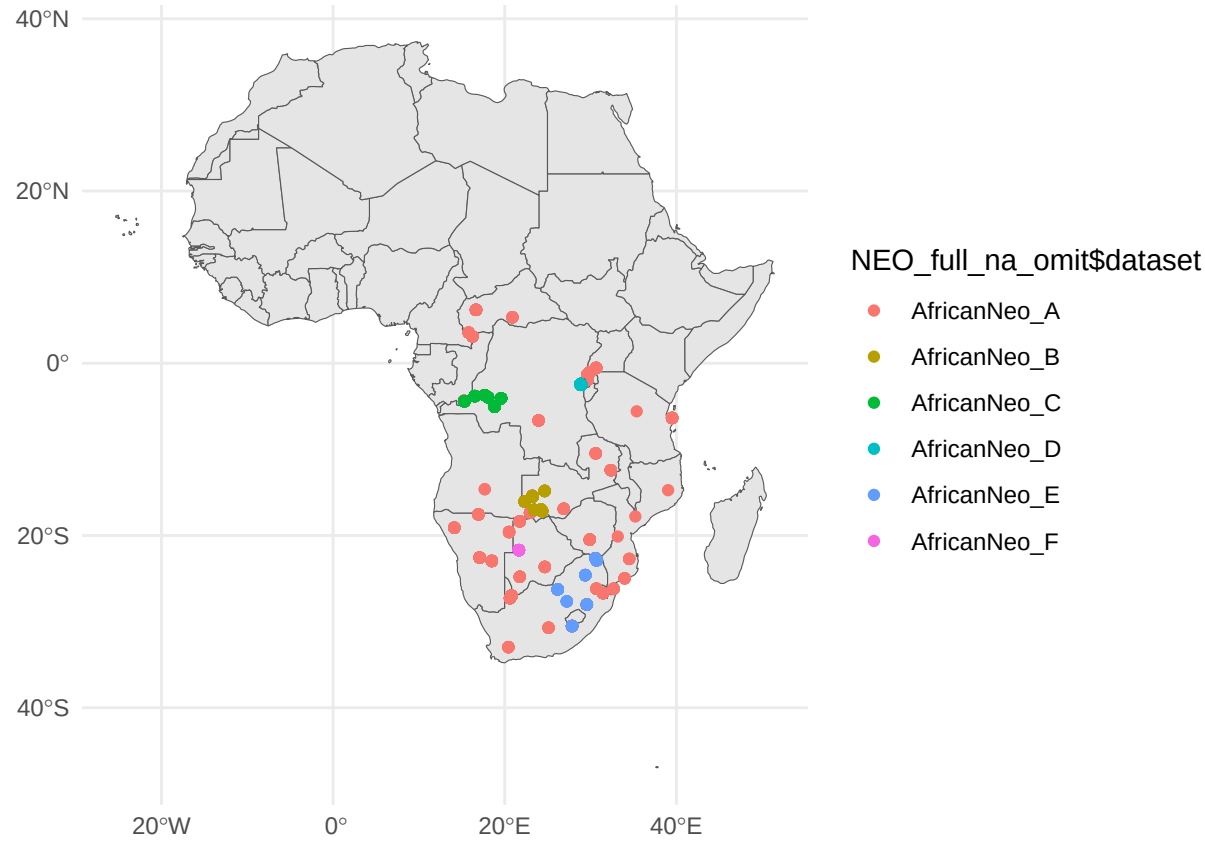
# add geometry to NEO_full_na_omit based on matching population labels
NEO_full_na_omit <- NEO_full_na_omit %>%
  left_join(all_pops,
    by = c("population" = "Population")
  )
```

```
## Warning in left_join(., all_pops, by = c(population = "Population")): Detected an unexpected many-to
## i Row 107 of `x` matches multiple rows in `y`.
## i Row 55 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```
NEO_full_na_omit <- gt_add_sf(NEO_full_na_omit,
  coords = c("Longitude", "Latitude"), crs = st_crs(4326)
)


africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")

ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = NEO_full_na_omit$geometry, aes(colour = NEO_full_na_omit$dataset)) +
  coord_sf(crs = st_crs(4326)) +
  theme_minimal()
```



Read in and clean up metadata

Using population metadata provided in the supplementary of Fortez-Lima et al., 2024.

```r
dat <- as.data.frame(read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 2))
```

```
## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
```

```r
colnames(dat) <- dat[1, ]

# rename the 'Only-Africa dataset' column
colnames(dat)[which(colnames(dat) == "Only-Africa dataset")] <- "Africa"

# include only entries in Africa
dat <- dat %>%
  filter(Africa == "Yes")

# remove rows with missing lat/lon
dat <- dat %>%
  filter(!is.na(Latitude) & !is.na(Longitude))

# transform to sf object
dat <- st_as_sf(dat, coords = c("Longitude", "Latitude"), crs = 4326)

# make sample size numeric
dat$N <- as.numeric(dat$N)
```

```r
# dat2 contains all populations in EGAS50000000006, even those without coordinates
dat2 <- read_excel("../data/raw/data_input/41586_2023_6770_MOESM4_ESM.xlsx", sheet = 1)
```

```
## New names:
## * `` -> `...1`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
```

```r
colnames(dat2) <- dat2[1,]
dat2 <- dat2[-1,]
```

```r
# rename the 'Population label.1' column
colnames(dat2)[8] <- "Population"

# not all have coordinates
setdiff(dat$`Population label`, dat2$Population)
```

```
##  [1] "Cameroon_Nzime"                 "Gabon_Nzebi"
##  [3] "Kenya_Luhya-LWK"                "Kenya_Kikuyu"
##  [5] "Kenya_Swahili-Mombasa"          "Kenya_Swahili-Kilifi"
##  [7] "Kenya_Swahili-Lamu"             "Uganda_Banyarwanda"
##  [9] "Uganda_Barundi"                 "Uganda_Baganda"
## [11] "Angola_Nyaneka"                 "Angola_Umbundu"
## [13] "Namibia_Damara-KSP"             "Mozambique_Yao"
## [15] "Mozambique_Nyanja"              "Mozambique_Tewe"
## [17] "Mozambique_Bitonga"             "SouthAfrica_SEBantu"
## [19] "SouthAfrica_SothoAGDP"          "SouthAfrica_ZuluAGDP"
## [21] "Gambia_Fula"                    "Gambia_Jola"
## [23] "Gambia_Mandinka"                "Gambia_Wolof"
## [25] "Gambia_Gambian-GWD"             "SierraLeone_Mende-MSL"
## [27] "Mali_Bwa"                       "IvoryCoast_Ahizi"
## [29] "IvoryCoast_Yacouba"             "Ghana_GaAdangbe"
## [31] "Benin_Bariba"                   "Benin_Fon"
## [33] "Benin_Yoruba"                   "Nigeria_Igbo"
## [35] "Nigeria_Esan-ESN"               "Nigeria_Yoruba-YRI"
## [37] "Ethiopia_Wolayta"               "Ethiopia_Amhara"
## [39] "Ethiopia_Oromo"                 "Ethiopia_Somali"
## [41] "Chad_Toubou"                    "Chad_Sara"
## [43] "Ethiopia_Gumuz"                 "Kenya_Kalenjin"
## [45] "Chad_Laal"                      "Cameroon_Baka"
## [47] "CameroonGabon_Baka"             "Ethiopia_Sabue"
## [49] "Tanzania_Hadza"                 "Namibia_Juhoansi"
## [51] "Botswana_KalahariKhoe"          "SouthAfrica_Coloured-Askham"
## [53] "SouthAfrica_Coloured-Colesberg" "SouthAfrica_Coloured-Wellington"
```

```r
length(unique(dat$`Population label`))
```

```
## [1] 111
```

```r
setdiff(NEO_full$population, dat$`Population label`)
```

```
##   [1] "Botswana_Kalanga"              "CAR_Nzakara"
##   [3] "CAR_Zande"                     "SouthAfrica_ColouredWellington"
##   [5] "SouthAfrica_Swazi"             "SouthAfrica_ColouredColesberg"
##   [7] "Namibia_Damara"                "Botswana_Kgalagadi"
##   [9] "SouthAfrica_SothoSouthern"     "SouthAfrica_Hlubi"
##  [11] "SouthAfrica_Mputhi"            "Mozambique_Nhungwe"
##  [13] "Mozambique_Tsonga"             "Mozambique_Ronga"
##  [15] "Mozambique_Chuwabu"            "Mozambique_TongaMoz"
##  [17] "SouthAfrica_Ndebele"           "SouthAfrica_SothoNorthern"
##  [19] "Botswana_TswanaBot"            "Lesotho_SothoSouthern"
##  [21] "SouthAfrica_Soto"              "DRC_Yombe"
##  [23] "DRC_Ngbandi"                   "DRC_Boo"
```

```
##  [25] "DRC_Mboma"                    "DRC_Ndibu"
##  [27] "DRC_MongoNkundo"              "DRC_Ntandu"
##  [29] "DRC_Mbunda"                   "DRC_Ngombe"
##  [31] "DRC_Tetela"                   "DRC_Sakata"
##  [33] "DRC_Nandi"                    "DRC_Ruanda"
##  [35] "DRC_Yaka"                     "DRC_Hendo"
##  [37] "DRC_Haavu"                    "DRC_BanguBangu"
##  [39] "DRC_Kanyoka"                  "DRC_Ciokwe"
##  [41] "DRC_Suku"                     "DRC_Kusu"
##  [43] "DRC_Songe"                    "DRC_Sengele"
##  [45] "DRC_Kango"                    "DRC_Kuba"
##  [47] "DRC_Zimba"                    "DRC_Lunda"
##  [49] "DRC_Binza"                    "Uganda_Konzo"
##  [51] "Zambia_Kaonde"                "Zambia_ShonaZam"
##  [53] "Zambia_Tumbuka"               "Zambia_Ngoni"
##  [55] "Zambia_Nsenga"                "SouthAfrica_Coloured"
##  [57] "Angola_Vasekele"              "Angola_Sekele"
##  [59] "Namibia_Kgalagadi"            "DRC_Luozi"
##  [61] "DRC_Nienge"                   "DRC_Bandundu"
##  [63] "Tanzania_Mixed"               "Zambia_Tswana"
##  [65] "Zambia_Kalanga"               "Zambia_Kgalagadi"
##  [67] "Zambia_Mbukushu"              "Zambia_Kavango"
##  [69] "Zambia_Kwangari"              "Zambia_Luyana"
##  [71] "Zambia_Shona"                 "Zambia_Shanjo"
##  [73] "DRC_Wonk"                     "DRC_Nsong"
##  [75] "DRC_Nzadi"                    "DRC_Hungan"
##  [77] "DRC_Nsambaan"                 "DRC_Lele"
##  [79] "DRC_MbweneNgong"              "DRC_Mpiin"
##  [81] "DRC_LubaKasai"                "DRC_Ngong"
##  [83] "DRC_Mpur"                     "DRC_Cokwe"
##  [85] "DRC_Mpe"                      "DRC_NunuBobangi"
##  [87] "DRC_NunuMushie"               "DRC_Ngbaka"
##  [89] "DRC_Balobo"                   "DRC_Mbudza"
##  [91] "DRC_Mongo"                    "DRC_Nande"
##  [93] "DRC_Lulua"                    "DRC_Binji"
##  [95] "DRC_Ruund"                    "DRC_LubaKatanga"
##  [97] "DRC_Mangbetu"                 "DRC_Mbaama"
##  [99] "DRC_Myene"                    "DRC_Nzebi"
## [101] "DRC_Teke"                     "DRC_Fang"
## [103] "DRC_RundiHutu"                "DRC_Lika"
## [105] "DRC_Luba"                     "SouthAfrica_Shona"
```

```
as.integer(table(NEO_full$population))
```

```
##   [1]  17  10   5   2  29  11   3   6   7  10  22  24  11   8   6   1   1   4
##  [19]   1   1   1   1   1  47   1   1   1   3   1   1   4   1   5   1   1   2
##  [37]   2  14   1   2   3  18   1  41   1  22   1   2   1  34   1   1   9   4
##  [55]   2   1   1   1   2   8   1   2   4   4  69   1   1   2   6   3   2   7
##  [73]   1  24  15   1   1   1   5   1 135   3   3   1   3   4   5  29   9   1
##  [91]   8   7   4   1   1   3   3   1   4   7   5  23  24  23  10  16  11  23
## [109]  25  13  12  14  19   3  10  32   3   8  28   1   7  20  29   1   6  32
## [127]  32  24  32  63  17   1  14  21  22   6  22  13  18  14  13   5   1  13
## [145]  16   1  10  25   5   9  27   8  16   5  10  10   1   4  10   2   5  23
## [163]  40
```

```
as.integer(na.omit(dat2$N))
```

```
##   [1]   17   10    5    2   29   11    3    6    7   10   22   24   11    8    6
##  [16]    1    1    4    1    1    1    1    1   47    1    1    1    3    1    1
##  [31]    4    1    5    1    1    2    2   14    1    2    3   18    1   41    1
##  [46]   22    1    2    1   34    1    1    9    4    2    1    1    1    2    8
##  [61]    1    2    4    4   69    1    1    2    6    3    2    7    1   24   15
##  [76]    1    1    1    5    1  135    3    3    1    3    4    5   29    9    1
##  [91]    8    7    4    1    1    3    3    1    4    7    5   23   24   23   10
## [106]   16   11   23   25   13   12   14   19    3   10   32    3    8   28    1
## [121]    7   20   29    1    6   32   32   24   32   63   17    1   14   21   22
## [136]    6   22   13   18   14   13    5    1   13   16    1   10   25    5    9
## [151]   27    8   16    5   10   10    1    4   10    2    5   23   40 1763
```

```
# The total number of individuals in the metadata is way larger than the number we have
sum(dat$N)
```

```
## [1] 3902
```

```
nrow(NEO_full)
```

```
## [1] 1763
```

Unfortunately, coordinates are missing for a lot of populations in our full dataset. In some cases the mismatch is a naming issue

eg

Namibia_DamaraKSP in dat$`Population label` is probably "Namibia_Damara" in NEO_full

There are coords for Sotho, but in NEO_full there are southern and northern Sotho

SouthAfrica_Coloured in NEO_full is probably SouthAfrica_ColouredAskham in dat$`Population label` - both have 12 individuals

```
dat$`Population label` <- ifelse(dat$`Population label` == "Tanzania_TanzaniaMixed",
  "Tanzania_Mixed",
  dat$`Population label`
)


# rename DRC_Bangubangu to DRC_BanguBangu in NEO_full$population
NEO_full$population <- ifelse(NEO_full$population == "DRC_Bangubangu",
  "DRC_BanguBangu",
  NEO_full$population
)


setdiff(NEO_full$population, dat$`Population label`)
```

```
##   [1] "Botswana_Kalanga"              "CAR_Nzakara"
##   [3] "CAR_Zande"                     "SouthAfrica_ColouredWellington"
##   [5] "SouthAfrica_Swazi"             "SouthAfrica_ColouredColesberg"
```

```
##   [7] "Namibia_Damara"                "Botswana_Kgalagadi"
##   [9] "SouthAfrica_SothoSouthern"     "SouthAfrica_Hlubi"
##  [11] "SouthAfrica_Mputhi"            "Mozambique_Nhungwe"
##  [13] "Mozambique_Tsonga"             "Mozambique_Ronga"
##  [15] "Mozambique_Chuwabu"            "Mozambique_TongaMoz"
##  [17] "SouthAfrica_Ndebele"           "SouthAfrica_SothoNorthern"
##  [19] "Botswana_TswanaBot"            "Lesotho_SothoSouthern"
##  [21] "SouthAfrica_Soto"              "DRC_Yombe"
##  [23] "DRC_Ngbandi"                   "DRC_Boo"
##  [25] "DRC_Mboma"                     "DRC_Ndibu"
##  [27] "DRC_MongoNkundo"               "DRC_Ntandu"
##  [29] "DRC_Mbunda"                    "DRC_Ngombe"
##  [31] "DRC_Tetela"                    "DRC_Sakata"
##  [33] "DRC_Nandi"                     "DRC_Ruanda"
##  [35] "DRC_Yaka"                      "DRC_Hendo"
##  [37] "DRC_Haavu"                     "DRC_BanguBangu"
##  [39] "DRC_Kanyoka"                   "DRC_Ciokwe"
##  [41] "DRC_Suku"                      "DRC_Kusu"
##  [43] "DRC_Songe"                     "DRC_Sengele"
##  [45] "DRC_Kango"                     "DRC_Kuba"
##  [47] "DRC_Zimba"                     "DRC_Lunda"
##  [49] "DRC_Binza"                     "Uganda_Konzo"
##  [51] "Zambia_Kaonde"                 "Zambia_ShonaZam"
##  [53] "Zambia_Tumbuka"                "Tanzania_TanzaniaMixed"
##  [55] "Zambia_Ngoni"                  "Zambia_Nsenga"
##  [57] "SouthAfrica_Coloured"          "Angola_Vasekele"
##  [59] "Angola_Sekele"                 "Namibia_Kgalagadi"
##  [61] "DRC_Luozi"                     "DRC_Nienge"
##  [63] "DRC_Bandundu"                  "Zambia_Tswana"
##  [65] "Zambia_Kalanga"                "Zambia_Kgalagadi"
##  [67] "Zambia_Mbukushu"               "Zambia_Kavango"
##  [69] "Zambia_Kwangari"               "Zambia_Luyana"
##  [71] "Zambia_Shona"                  "Zambia_Shanjo"
##  [73] "DRC_Wonk"                      "DRC_Nsong"
##  [75] "DRC_Nzadi"                     "DRC_Hungan"
##  [77] "DRC_Nsambaan"                  "DRC_Lele"
##  [79] "DRC_MbweneNgong"               "DRC_Mpiin"
##  [81] "DRC_LubaKasai"                 "DRC_Ngong"
##  [83] "DRC_Mpur"                      "DRC_Cokwe"
##  [85] "DRC_Mpe"                       "DRC_NunuBobangi"
##  [87] "DRC_NunuMushie"                "DRC_Ngbaka"
##  [89] "DRC_Balobo"                    "DRC_Mbudza"
##  [91] "DRC_Mongo"                     "DRC_Nande"
##  [93] "DRC_Lulua"                     "DRC_Binji"
##  [95] "DRC_Ruund"                     "DRC_LubaKatanga"
##  [97] "DRC_Mangbetu"                  "DRC_Mbaama"
##  [99] "DRC_Myene"                     "DRC_Nzebi"
## [101] "DRC_Teke"                      "DRC_Fang"
## [103] "DRC_RundiHutu"                 "DRC_Lika"
## [105] "DRC_Luba"                      "SouthAfrica_Shona"
```

```r
# add coordinates to NEO_full based on matching population labels
NEO_full_na_omit <- NEO_full %>%
  filter(population %in% dat$`Population label`)
```

```r
dat_sep <- dat %>%
  mutate(lat = st_coordinates(.)[,1],
         long = st_coordinates(.)[,2]) %>%
  st_drop_geometry()

# rename `population label` to population in dat_sep
colnames(dat_sep)[which(colnames(dat_sep) == "Population label")] <- "population"


# add geometry to NEO_full_na_omit based on matching population labels
NEO_full_na_omit <- NEO_full_na_omit %>%
  left_join(dat_sep[, c("population", "lat", "long")],
    by = c("population")
  )


NEO_full_na_omit <- gt_add_sf(NEO_full_na_omit,
  coords = c("lat", "long"), crs = st_crs(4326)
)
```

```r
africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")

ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = NEO_full_na_omit$geometry, aes(colour = NEO_full_na_omit$population)) +
  coord_sf(crs = st_crs(4326)) +
  theme_minimal()
```




```
setdiff(NEO_full$population, NEO_full_na_omit$population)
```

```
##   [1] "Botswana_Kalanga"              "CAR_Nzakara"
##   [3] "CAR_Zande"                     "SouthAfrica_ColouredWellington"
##   [5] "SouthAfrica_Swazi"             "SouthAfrica_ColouredColesberg"
##   [7] "Namibia_Damara"                "Botswana_Kgalagadi"
##   [9] "SouthAfrica_SothoSouthern"     "SouthAfrica_Hlubi"
##  [11] "SouthAfrica_Mputhi"            "Mozambique_Nhungwe"
##  [13] "Mozambique_Tsonga"             "Mozambique_Ronga"
##  [15] "Mozambique_Chuwabu"            "Mozambique_TongaMoz"
##  [17] "SouthAfrica_Ndebele"           "SouthAfrica_SothoNorthern"
##  [19] "Botswana_TswanaBot"            "Lesotho_SothoSouthern"
##  [21] "SouthAfrica_Soto"              "DRC_Yombe"
##  [23] "DRC_Ngbandi"                   "DRC_Boo"
##  [25] "DRC_Mboma"                     "DRC_Ndibu"
##  [27] "DRC_MongoNkundo"               "DRC_Ntandu"
##  [29] "DRC_Mbunda"                    "DRC_Ngombe"
##  [31] "DRC_Tetela"                    "DRC_Sakata"
##  [33] "DRC_Nandi"                     "DRC_Ruanda"
##  [35] "DRC_Yaka"                      "DRC_Hendo"
##  [37] "DRC_Haavu"                     "DRC_BanguBangu"
##  [39] "DRC_Kanyoka"                   "DRC_Ciokwe"
##  [41] "DRC_Suku"                      "DRC_Kusu"
##  [43] "DRC_Songe"                     "DRC_Sengele"
##  [45] "DRC_Kango"                     "DRC_Kuba"
```

```
##  [47] "DRC_Zimba"                    "DRC_Lunda"
##  [49] "DRC_Binza"                    "Uganda_Konzo"
##  [51] "Zambia_Kaonde"                "Zambia_ShonaZam"
##  [53] "Zambia_Tumbuka"               "Tanzania_TanzaniaMixed"
##  [55] "Zambia_Ngoni"                 "Zambia_Nsenga"
##  [57] "SouthAfrica_Coloured"         "Angola_Vasekele"
##  [59] "Angola_Sekele"                "Namibia_Kgalagadi"
##  [61] "DRC_Luozi"                    "DRC_Nienge"
##  [63] "DRC_Bandundu"                 "Zambia_Tswana"
##  [65] "Zambia_Kalanga"               "Zambia_Kgalagadi"
##  [67] "Zambia_Mbukushu"              "Zambia_Kavango"
##  [69] "Zambia_Kwangari"              "Zambia_Luyana"
##  [71] "Zambia_Shona"                 "Zambia_Shanjo"
##  [73] "DRC_Wonk"                     "DRC_Nsong"
##  [75] "DRC_Nzadi"                    "DRC_Hungan"
##  [77] "DRC_Nsambaan"                 "DRC_Lele"
##  [79] "DRC_MbweneNgong"              "DRC_Mpiin"
##  [81] "DRC_LubaKasai"                "DRC_Ngong"
##  [83] "DRC_Mpur"                     "DRC_Cokwe"
##  [85] "DRC_Mpe"                      "DRC_NunuBobangi"
##  [87] "DRC_NunuMushie"               "DRC_Ngbaka"
##  [89] "DRC_Balobo"                   "DRC_Mbudza"
##  [91] "DRC_Mongo"                    "DRC_Nande"
##  [93] "DRC_Lulua"                    "DRC_Binji"
##  [95] "DRC_Ruund"                    "DRC_LubaKatanga"
##  [97] "DRC_Mangbetu"                 "DRC_Mbaama"
##  [99] "DRC_Myene"                    "DRC_Nzebi"
## [101] "DRC_Teke"                     "DRC_Fang"
## [103] "DRC_RundiHutu"                "DRC_Lika"
## [105] "DRC_Luba"                     "SouthAfrica_Shona"
```

```
View(dat[which(dat$Country == "Angola"),])
```

DRC_LubaLulua has 11 indivs, NEO_full has 1 indiv for DRC_Lulua

```
length(unique(NEO_full$population))
```

```
## [1] 163
```

```
length(unique(NEO_full_na_omit$population))
```

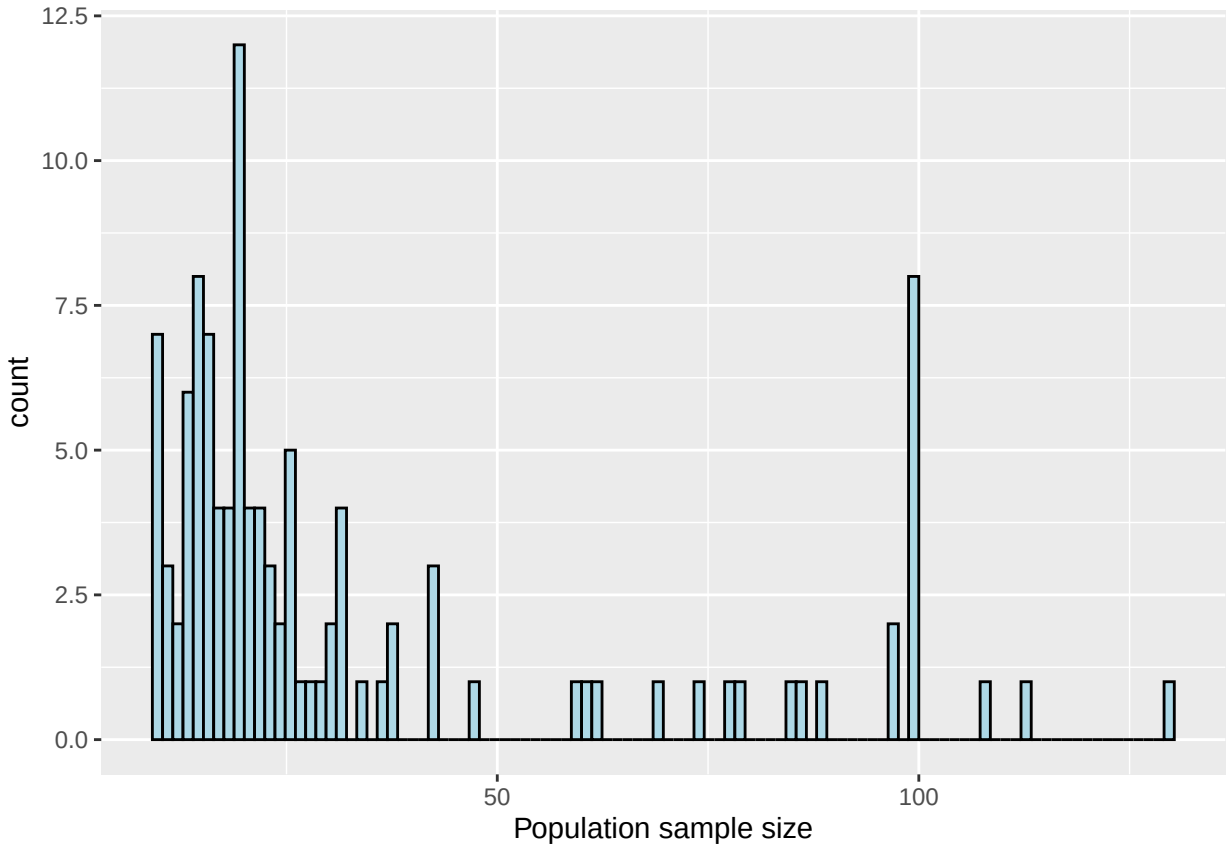
```
## [1] 57
```

```
length(unique(dat$`Population label`))
```

```
## [1] 111
```

Current population sample sizes

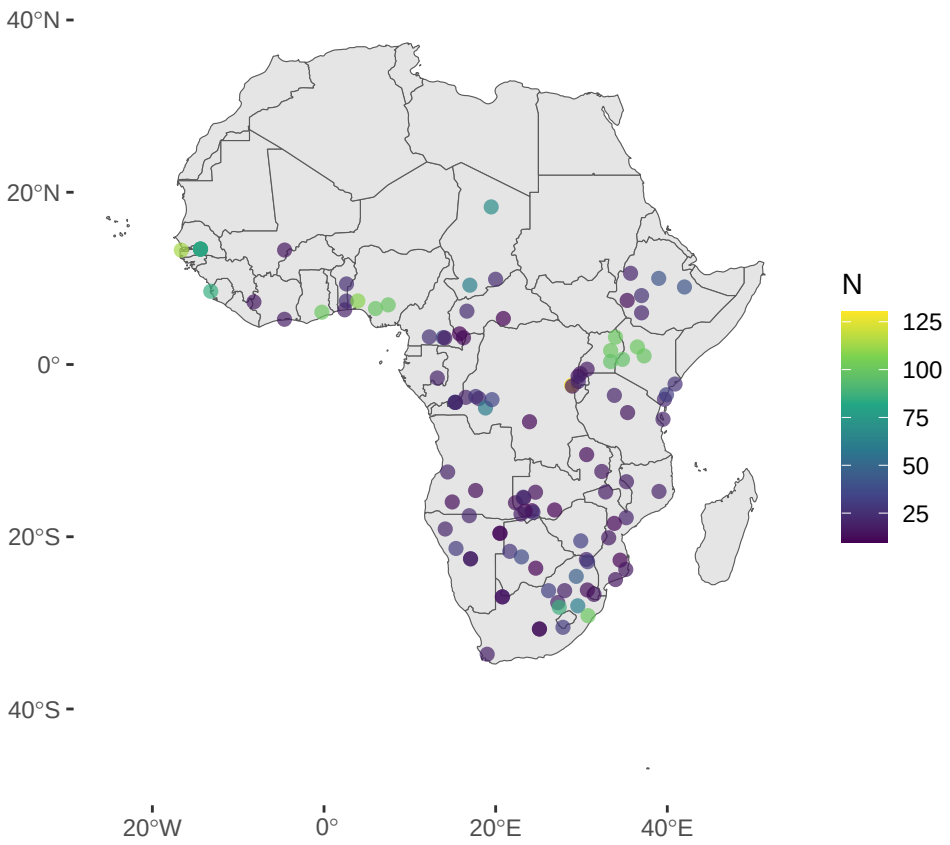
```
ggplot(dat) +
  geom_histogram(aes(x = N), bins = 100, fill = "lightblue", color = "black") +
  labs(x = "Population sample size")
```



After filtering, some populations contain very few individuals. This causes an issue for calculating F statistics between populations.

```
africa <- ne_countries(continent = c("Africa"), scale = 50, returnclass = "sf")

p2 <- ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat, aes(color = N), size = 2, alpha = 0.7) +
  theme(panel.background = element_rect(fill = "white", color = NA)) +
  scale_color_viridis_c()
p2
```



Potential alternative groupings

- Glottocode and Guthrie group are not available for all populations, don't group on this basis.

A few options:

1. Group by linguistic group

For the Bantu speaking groups, each population is given a 'branch', and for the non-Bantu groups, we could use prexisting groups. This results in very unbalanced sample sizes (highest is Eastern Bantu-speaking at 1580 individuals and lowest is 23 individuals). This will also put the Mozambique individuals in a much larger Eastern Bantu-speaking group, which makes our result less specific.

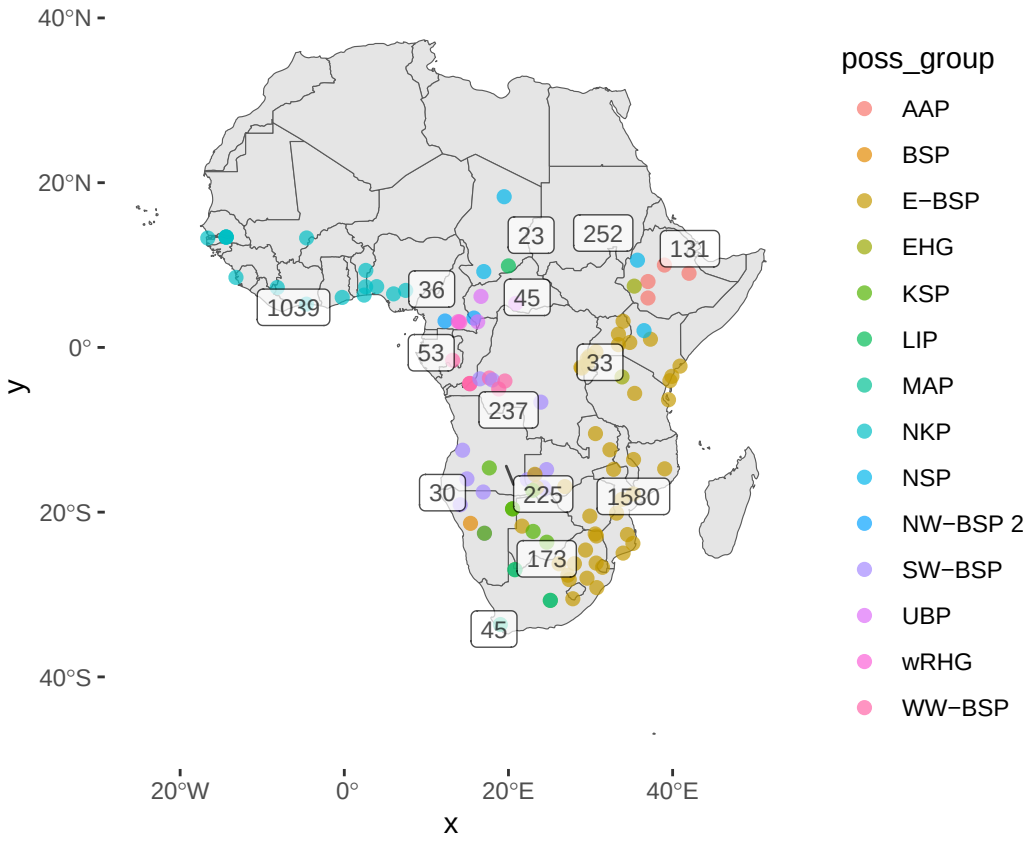
```
dat$`Bantu Group` <- ifelse(dat$`Bantu Group` == "-", NA, dat$`Bantu Group`)
dat$poss_group <- ifelse(!is.na(dat$`Bantu Group`), dat$`Bantu Group`, dat$Group)
table(dat$poss_group)
```

```
##
##      AAP      BSP    E-BSP      EHG      KSP      LIP      MAP      NKP
##        4        1       44        2        9        1        3       16
##      NSP NW-BSP 2   SW-BSP      UBP    wRHG   WW-BSP
##        4        2       13        3        2        7
```

```
label_data <- dat %>%
  group_by(poss_group) %>%
  summarise(n = sum(N), geometry = st_centroid(st_union(geometry))) %>%
  arrange(-desc(n)) # Results in 14 groups
```

```
ggplot() +
  geom_sf(data = africa) +
  geom_sf(data = dat, aes(color = poss_group), size = 2, alpha = 0.7) +
  geom_label_repel(
    data = label_data, force = 5, force_pull = 1, aes(label = n, geometry = geometry),
    stat = "sf_coordinates", size = 3, color = "black", alpha = 0.7
  ) +
  theme(panel.background = element_rect(fill = "white", color = NA))
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```



2. Use linguistic groups to guide cases where we merge populations together

eg if a set of populations have less than X individuals each, are part of the same linguistic branch, and within X distance of one another then we put them together.

If a population has 15 or fewer individuals, we look for other populations within 500km that are part of the same linguistic group and save them as potential merge candidates

```r
distance_matrix <- st_distance(dat)
rownames(distance_matrix) <- dat$Population
colnames(distance_matrix) <- dat$Population
distance_matrix <- distance_matrix / 1000


merge_pop_names <- function(x, distance_matrix, dat) {
  # if n <= 10
  if (as.integer(x["N"]) <= 15) {
    # get the name of the population
    pop_name <- x["Population"]
    # get the group of the population
    group <- x["poss_group"]
    # find this row in the distance matrix
    dists <- distance_matrix[pop_name, ]
    # drop units
    dists <- units::drop_units(dists)
    # get the names of populations within 500km
    close_pops <- names(dists[dists <= 500])
    # exclude self-distance
    close_pops <- close_pops[close_pops != pop_name]
    # get the group of populations within 200km
    close_pops_groups <- dat$poss_group[dat$Population %in% close_pops]
    # if any pops are closer than 200 km, and within the same poss_group
    if (length(close_pops_groups > 0) & any(close_pops_groups == group)) {
      # return the names of the populations to merge
      return(paste(c(pop_name, close_pops[close_pops_groups == group]), collapse = "; "))
    } else {
      # if there's nothing it can be merged with within 200km
      return("Nothing to merge")
    }
  } else if (x["N"] > 10) {
    return(NA)
  }
}

dat$merge_names <- apply(st_drop_geometry(dat), 1, merge_pop_names,
  distance_matrix = distance_matrix, dat = dat
)

# NB consider two Baka entries with the same Population and Group
```

Run admixture on only Eastern Bantu pops of interest

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.